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1 Goal vs Cost: Goal-based achieves specified goal; Cost-based minimizes expected long-term cost by weighting different errors.
2 PEAS: Performance, Environment, Actuators, Sensors. Used to specify an agent's task environment.
3 Capacities: Sense→Reason→Act→Learn using feedback.

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1 Autonomous taxi: Mainly multi-agent; both cooperative and competitive.
2 Poker: partially observable, strategic, sequential, dynamic, discrete, multi-agent. Medical diagnosis: partially observable, stochastic, sequential, dynamic, continuous, single-agent. Image analysis: fully/partially observable, deterministic, episodic, static, single-agent. Interactive English tutor: partially observable, stochastic, sequential, dynamic, multi-agent.
3 Known vs unknown is different from fully vs partially observable. Known refers to knowing rules; observable refers to access to state.
4 Episodic easier because each decision independent. Communication emerges in multi-agent settings to coordinate and improve outcomes.

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1 Data Science: extracting knowledge using statistics, ML and computing. Learning agent: improves from experience. ML: algorithms that learn patterns.
2 5Vs: Volume, Velocity, Variety, Veracity, Value.
3 ML vs Data Mining: prediction vs discovering unknown patterns. ML vs DL: DL is subset of ML using deep neural networks.
4 Supervised ML: Training Data->Learning Algorithm->Model->Prediction.
5 Life expectancy: Regression. Customer segmentation: Clustering. Robot navigation: Reinforcement Learning. Recommender: Recommendation/collaborative filtering. Weather forecasting: Regression/Time-series. Medical image classification: Classification.